

DAY 14: Scope of variables

1. Local variable

→ A variable that is define inside the function call it as a local variable, where the variable can only access with in that function

Eg:

```
def display():  
    name='teja' #local variable  
    print(name)  
  
display()
```

2. Global variable

→ A variable that is define outside the function call it as global variable and it can be access anywhere thorough out program

Eg:

```
a=90 #global variable  
  
def display():  
    print(a)  
  
display()
```

Global keyword

→ global is keyword used to re-access new values to variable that was already define outside the function call

Eg:

```
a=90 #global variable  
print(a)  
  
def display():  
    global a  
    a=50  
  
display()  
print(a)
```

passing by value

```
def even_odd(num):  
    if num%2==0:  
        print(f'{num} is even')  
    else:  
        print(f'{num} is odd')  
even_odd(90) # passing by value
```

passing by reference

```
num=9 # reference
def even_odd(num):
    if num%2==0:
        print(f'{num} is even')
    else:
        print(f'{num} is odd')
even_odd(num)
```

Recursive function

→ The function call itself until the base condition met...

Eg:

```
def fac(a):
    if a==0 or a==1:
        return a
    return a*fac(a-1)
print(fac(5))
```